

Figure 6: Reinforcement Learning results for 5 DeepMind Lab tasks used in [50]. Black: batched A2C baseline, Red: with auxiliary contrastive loss.

3.4 Reinforcement Learning

Finally, we evaluate the proposed unsupervised learning approach on five reinforcement learning in 3D environments of DeepMind Lab [51]: rooms_watermaze, explore_goal_locations_small, seekavoid_arena_01, lasertag_three_opponents_small and rooms_keys_doors_puzzle.

This setup differs from the previous three. Here, we take the standard batched A2C [52] agent as base model and add CPC as an auxiliary loss. We do not use a replay buffer, so the predictions have to adapt to the changing behavior of the policy. The learned representation encodes a distribution over its future observations.

Following the same approach as [50], we perform a random search over the entropy regularization weight, the learning-rate and epsilon hyperparameters for RMSProp [53]. The unroll length for the A2C is 100 steps and we predict up to 30 steps in the future to derive the contrastive loss. The baseline agent consists of a convolutional encoder which maps every input frame into a single vector followed by a temporal LSTM. We use the same encoder as in the baseline agent and only add the linear prediction mappings for the contrastive loss, resulting in minimal overhead which also showcases the simplicity of implementing our method on top of an existing architecture that has been designed and tuned for a particular task. We refer to [50] for all other hyperparameter and implementation details.

Figure 6 shows that for 4 out of the 5 games performance of the agent improves significantly with the contrastive loss after training on 1 billion frames. For lasertag_three_opponents_small, contrastive loss does not help nor hurt. We suspect that this is due to the task design, which does not require memory and thus yields a purely reactive policy.

4 Conclusion

In this paper we presented Contrastive Predictive Coding (CPC), a framework for extracting compact latent representations to encode predictions over future observations. CPC combines autoregressive modeling and noise-contrastive estimation with intuitions from predictive coding to learn abstract representations in an unsupervised fashion. We tested these representations in a wide variety of domains: audio, images, natural language and reinforcement learning and achieve strong or state-of-the-art performance when used as stand-alone features. The simplicity and low computational requirements to train the model, together with the encouraging results in challenging reinforcement learning domains when used in conjunction with the main loss are exciting developments towards useful unsupervised learning that applies universally to many more data modalities.

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